

Pseudoparticles and confinement in the two-dimensional Abelian Higgs model

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Following Polyakov's proposal that certain classical trajectories (pseudoparticles) can restore broken symmetries and provide a mechanism for quark confinement, we study the Abelian Higgs model and calculate the pseudoparticle (vortex) contribution to functional integrals, taking account of the interaction between vortices. We obtain confinement in the sense of Wilson for fractionally charged quarks.

I. INTRODUCTION

Polyakov¹ has pointed out that Euclidean solutions of gauge theories (pseudoparticles) may dominate the functional integral which describes a quantum field theory continued to Euclidean space.

It was shown^{2,3} that, for a Yang-Mills theory, the classical vacuums are degenerate and pseudoparticle configurations represent a tunneling phenomenon between the vacuums. When the correct vacuum is constructed as a coherent superposition of the classical vacuums, it is realized that in the new vacuum, labeled by a parameter θ ($|\theta| < \pi$), fermionic symmetries are affected.^{2,3,4}

In two space-time dimensions there is a theory with well-known pseudoparticle solutions: the Abelian Higgs model. This model has vortex-type classical solutions as has been put forward by Nielsen and Olesen.⁵ Recently, Callan, Dashen, and Gross⁶ have discussed this model in the dilute-gas approximation. Neglecting interactions between vortices, they showed that external charged particles are confined only if q/e is not an integer (q is the charge of the external particle, e is the fundamental charge). It is the purpose of this note to take account of the interaction between vortices and calculate the force between two external heavy charged particles, following a method developed by Polyakov.⁷ In Sec. II we compute the energy interaction between vortices in analogy with what is done in the Ginzburg-Landau theory of superconductivity. After eliminating zero-energy modes of the quantum fluctuations (Sec. III), we discuss vortex contributions to the functional integral in the one-loop approximation and construct some relevant correlation functions (Sec. IV).

Finally, in Sec. V, we study the problem of confinement of very heavy charged particles introduced in the theory. We explicitly show that, when $q/e = \text{semi-integer}$, the force between the charges is independent of the distance, thus confirming the result advanced by Callan, Dashen, and Gross.

II. THE ABELIAN HIGGS MODEL

We consider the Lagrangian density of a Higgs model:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}(\partial_\mu + ieA_\mu)\phi|^2 + c_2|\phi|^2 - c_4|\phi|^4, \quad (2.1)$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (2.2)$$

Nielsen and Olesen⁵ found vortex-type static solutions of the equations of motion derived from the Lagrangian (2.1). The flux of vortex lines is quantized in units of $2\pi/e$. The solution corresponding to n units of flux has the form

$$\begin{aligned} \phi(\vec{r}) &= e^{-in\varphi} f(\rho), \\ \vec{A}(\vec{r}) &= -\hat{e}_\varphi \frac{A(\rho)}{\rho}, \end{aligned} \quad (2.3)$$

where (ρ, φ) are polar coordinates. The fields defined in Eqs. (2.3) can also be considered as the solution of Euclidean field theory defined by a Lagrangian density derived from Eq. (2.1), in two space-time dimensions.

The appropriate boundary conditions for $f(\rho)$ and $A(\rho)$ are

$$\begin{aligned} f(0) &= 0, \quad \lim_{\rho \rightarrow \infty} f(\rho) = \left(\frac{c_2}{2c_4}\right)^{1/2} = f_0, \\ A(0) &= 0, \quad \lim_{\rho \rightarrow \infty} A(\rho) = -\frac{n}{e}. \end{aligned} \quad (2.4)$$

Hence, in rectangular components the vector field reads asymptotically

$$\vec{A}_{\text{asy}} = \frac{n}{e} \left(\frac{y}{\rho^2}, -\frac{x}{\rho^2} \right), \quad (2.5)$$

while the real and imaginary part of the Higgs field take the form (in the case $|n|=1$)

$$\phi_1 \rightarrow f_0 \frac{x}{\rho}, \quad \phi_2 \rightarrow (-1)^{(n+1)/2} f_0 \frac{y}{\rho}. \quad (2.6)$$

The energy density can be written in the form (for the case $|n|=1$)

$$\epsilon = \frac{m_V^2}{e^2} Q\left(\frac{c_4}{e^2}\right), \quad (2.7)$$

where $m_V = e(c_2/2c_4)^{1/2}$ is the vector-meson mass. Function Q is slowly varying. It has been computed approximately by Neilsen and Olesen,

$$Q\left(\frac{c_4}{e^2}\right) \approx \frac{1}{2} \ln \frac{2(2c_4)^{1/2}}{e} + \pi \int_0^\infty K_0^2(y) y dy, \quad (2.8)$$

and exactly^{8,9} in the particular case $e^2 = 8c_4$, which corresponds, if one identifies the theory defined by Lagrangian (2.1) with the Ginzburg-Landau model of superconductivity, to the limiting case between type-II and type-I superconductivity. In this particular case

$$Q\left(\frac{c_4}{e^2} = \frac{1}{8}\right) = \pi. \quad (2.9)$$

For arbitrary n , one can easily see that, excepting the limiting case $e^2 = 8c_4$, the energy of one vortex is a quadratic function of the flux. Then, it is preferable to have two vortices of flux ϕ^0 (total energy $2\epsilon^{(1)}$) than one vortex of double flux (energy $4\epsilon^{(1)}$). This will justify the choice of minimum flux value ($|n|=1$) in the following calculations.

Let us consider now the case of several vortices with $n = \pm 1$, fixed at certain points $\{r_a\}$ [$r_a = (x_a, y_a)$] far from each other. This corresponds to the case

$$|\vec{r}_a - \vec{r}_b| \gg 1/m_V, \quad (2.10)$$

since for $m_S \sim m_V$ (Ref. 5) we have a well-defined vortex line of extension $\sim 1/m_V$. Provided one is far enough from each vortex, the solution can be written as a superposition of fields:

$$\phi_1 = \sum_a \frac{f_0}{|\vec{r} - \vec{r}_a|} (x - x_a), \quad (2.11)$$

$$\phi_2 = \sum_a (-1)^{(n_a+1)/2} \frac{f_0}{|\vec{r} - \vec{r}_a|} (y - y_a),$$

$$A_x = \frac{1}{e} \sum_a n_a \frac{y - y_a}{|\vec{r} - \vec{r}_a|^2}, \quad (2.12)$$

$$A_y = -\frac{1}{e} \sum_a n_a \frac{x - x_a}{|\vec{r} - \vec{r}_a|^2}.$$

The energy (action) at this extremum can be calculated in the following way: One surrounds each vortex with a circle of radius R such that $R \gg 1/m_V$ but $R \ll |\vec{r}_a - \vec{r}_b| = R_{ab}$. Inside each circle solutions (2.11), (2.12) are no more valid but, because $R_{ab} \gg R$, one can neglect the influence of all the other vortices on a given one.

Then, the first contribution to the energy, $\epsilon^{(1)}$, corresponding to regions inside the circles is calculated

simply as the sum of individual vortex energies:

$$\epsilon^{(1)} = \sum_a \frac{m_V^2}{e^2} Q\left(\frac{c_4}{e^2}\right). \quad (2.13)$$

Outside the circles, because $R \ll R_{ab}$, asymptotic solutions (2.11), (2.12) can be used to evaluate

$$\epsilon^{(2)} = - \int_{\text{outside circles}} \mathcal{L} d^2x. \quad (2.14)$$

Instead of computing the integral (2.14) we will follow the procedure used in the Ginzburg-Landau theory of type-II superconductivity to evaluate the energy of a system of vortices.⁹ Let us define

$$J_\nu = -e^2 |\phi|^2 A_\nu - e |\phi|^2 \partial_\nu \chi, \quad (2.15)$$

$$\chi = -n\varphi. \quad (2.16)$$

Then, the equation of motion for the A field can be written

$$\partial_\mu F^{\mu\nu} = J^\nu. \quad (2.17)$$

Integrating J_ν around a contour C of large enough radius (so that the Higgs field takes its vacuum value), we have

$$\oint J_\nu dx^\nu = -e^2 f_0^2 \oint A_\nu dx^\nu - e f_0^2 \oint \partial_\nu \chi dx^\nu. \quad (2.18)$$

Using Eq. (2.16) and the fact that

$$\nabla \times \vec{A} = H \hat{k} \quad (2.19)$$

we have, for one vortex,

$$\int_{S_C} \nabla \times \vec{J} \Big|_k d^2x + e^2 f_0^2 \int_{S_C} H d^2x = 2n\pi e f_0^2. \quad (2.20)$$

Although Eq. (2.20) is valid outside the hard core (that is, in the region where $\phi \sim f_0$), if the hard-core radius is small enough (the scalar mass large enough) one can try to replace the corresponding singularity inside the hard core by a δ function. We can then write

$$\nabla \times \vec{J} + e^2 f_0^2 \vec{H} = 2n\pi e \phi_0^2 \delta^{(2)}(\vec{r}) \hat{k}. \quad (2.21)$$

Using the other Maxwell equation [Eq. (2.17)]

$$\nabla \times \vec{H} = \vec{J} \quad (2.22)$$

combined with Eq. (2.21), we finally have

$$\nabla \times (\nabla \times \vec{H}) + e^2 f_0^2 \vec{H} = 2n\pi f_0^2 e \delta^{(2)}(\vec{r}) \hat{k}. \quad (2.23)$$

This equation has the solution given by Nielsen and Olesen:

$$\vec{H} = \frac{m_V^2}{e} n K_0(m_V \rho) \hat{k}. \quad (2.24)$$

Now it is easy to modify Eq. (2.23) to the case when many parallel vortices are present:

$$\nabla \times (\nabla \times \vec{H}) + e^2 f_0^2 \vec{H} = 2\pi f_0^2 e \sum_a n_a \delta^{(2)}(\vec{r} - \vec{r}_a) \hat{k},$$

$$n_a = \pm 1, \quad (2.25)$$

with the corresponding solution

$$\vec{H} = \frac{m_V^2}{e} \sum_a n_a K_0(m_V |\vec{r} - \vec{r}_a|) \hat{k}. \quad (2.26)$$

We can then write for the energy

$$\epsilon = \int d^2x \left[(\nabla \times \vec{A})^2 + \frac{J^2}{e^2 f_0^2} \right]$$

$$= \frac{1}{2} \int d^2x \left[\vec{H} \cdot \vec{H} + \frac{1}{m_V^2} (\nabla \times \vec{H}) \cdot (\nabla \times \vec{H}) \right]. \quad (2.27)$$

Integrating by parts the second term in expression (2.27) and using Eqs. (2.25), (2.26), we find

$$\epsilon = \frac{\pi m_V^2}{e^2} \sum_{a \neq b} n_a n_b K_0(m_V |\vec{r}_a - \vec{r}_b|) + \frac{\pi m_V^2}{e^2} \sum_{a=b} \int d^2x \delta^{(2)}(\vec{r}_a - \vec{r}_b) K_0(m_V |\vec{r}_a - \vec{r}_b|). \quad (2.28)$$

The logarithmic divergence in the second term reflects the singularity near the core. This integral must be cut off and its contribution is already taken into account in $\epsilon^{(1)}$. The first term corresponds to

$$\epsilon^{(2)} = \frac{\pi m_V^2}{e^2} \sum_{a \neq b} n_a n_b K_0(m_V |\vec{r}_a - \vec{r}_b|). \quad (2.29)$$

Then

$$\epsilon \cong \epsilon^{(1)} + \epsilon^{(2)} = \frac{m_V^2}{e^2} \left[\sum_a Q\left(\frac{c_4}{e^2}\right) + \pi \sum_{a \neq b} n_a n_b K_0(m_V |\vec{r}_a - \vec{r}_b|) \right]. \quad (2.30)$$

III. ZERO MODES

We shall consider the Euclidean formulation of the field theory, given the functional

$$S = \int d^2x \left[\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} |\nabla_\mu \phi|^2 + V(\phi) \right], \quad (3.1)$$

where

$$V(\phi) = -c_2 |\phi|^2 + c_4 (|\phi|^2)^2 + c_2^2 / 2c_4 \quad (3.2)$$

and

$$\nabla_\mu = \partial_\mu + ieA_\mu. \quad (3.3)$$

Let us consider a small deviation from the equilibrium:

$$\varphi = \varphi_{c1} + \phi, \quad (3.4)$$

$$A_\mu = A_\mu^{c1} + a_\mu,$$

where φ_{c1}, A_μ^{c1} are solutions of the classical equations of motion. We then obtain for functional (3.1) a quadratic form in ϕ and a_μ :

$$S = S_{c1} + S_{II}, \quad (3.5)$$

$$S_{II} = \int d^2x \left\{ \frac{1}{4} (\partial_\mu a_\nu - \partial_\nu a_\mu)^2 + \frac{1}{2} |\nabla_\mu^{c1} \phi|^2 + \frac{1}{2} ie a_\mu [\phi (\nabla_\mu^{c1} \varphi_{c1})^* - \phi^* (\nabla_\mu^{c1} \varphi_{c1}) + \varphi_{c1} (\nabla_\mu^{c1} \phi)^* - \varphi_{c1}^* (\nabla_\mu^{c1} \phi)] \right.$$

$$\left. + \frac{1}{2} e^2 a_\mu a^\mu |\varphi_{c1}|^2 + V_{II}(\varphi_{c1}, \phi) \right\}, \quad (3.6)$$

where

$$\nabla_\mu^{c1} = \partial_\mu + ieA_\mu^{c1} \quad (3.7)$$

and

$$V_{II} = -c_2 |\phi|^2 + 2c_4 |\varphi_{c1}|^2 |\phi|^2 + c_4 (\varphi_{c1} \phi^* + \varphi_{c1}^* \phi)^2. \quad (3.8)$$

In order to perform the functional integration we need eigenvalues and eigenfunctions of the quadratic form. They are defined by the equations

$$\partial^\mu (\partial_\mu a_\nu^{(n)} - \partial_\nu a_\mu^{(n)}) + ie [\phi^{(n)} (\nabla_\nu^{c1} \varphi_{c1})^* - \phi^{(n)*} (\nabla_\nu^{c1} \varphi_{c1}) + \varphi_{c1} (\nabla_\nu^{c1} \phi^{(n)})^* - \varphi_{c1}^* (\nabla_\nu^{c1} \phi^{(n)})] - e^2 a_\nu^{(n)} |\varphi_{c1}|^2 = -\Omega_n^2 a_\nu^{(n)} \quad (3.9)$$

and

$$\nabla_{\mu}^{c1}(\nabla^{\mu c1}\phi^{(n)}) + 2iea_{\mu}^{(n)}\nabla^{\mu c1}\varphi_{c1} + ie\varphi_{c1}\partial_{\mu}a^{\mu(n)} - 2\frac{\delta V_{II}}{\delta\phi^{(n)*}} = \Omega_n^2\phi^{(n)}, \quad (3.10)$$

where

$$\frac{\delta V_{II}}{\delta\phi^{(n)*}} = -c_2\phi^{(n)} + 2c_4|\varphi_{c1}|^2\phi^{(n)} + 2c_4\varphi_{c1}(\varphi_{c1}\phi^{(n)*} + \varphi_{c1}^*\phi^{(n)}).$$

As usual, we are faced with the problem of zero-mode eigenfunctions which introduce divergences that would render the functional integral infinite (functional integration in those directions is not Gaussian).

Let us first consider zero modes associated with gauge degrees of freedom. One can easily show that every eigenfunction of (3.9), (3.10) with nonzero eigenvalue ($\Omega_n^2 \neq 0$) satisfies automatically the condition

$$\frac{1}{e}\partial_{\mu}a_{\mu}^{(n)} + \varphi_1^{c1}\phi_2^{(n)} - \varphi_2^{c1}\phi_1^{(n)} = 0, \quad (3.11)$$

where $\varphi_1^{c1}, \varphi_2^{c1}$ ($\phi_1^{(n)}, \phi_2^{(n)}$) are the real and imaginary parts of φ^{c1} ($\phi^{(n)}$). The proof is as follows: Eqs. (3.9), (3.10) have zero-eigenvalue solutions given by

$$a_{\mu}^{(0)} = \frac{1}{e}\partial_{\mu}\alpha(x), \quad (3.12)$$

$$\phi^{(0)} = -i\alpha(x)\varphi^{c1},$$

since they correspond simply to infinitesimal gauge transformations of A_{μ}^{c1} and φ_{c1} which do not change S . Because zero-mode eigenfunctions are orthogonal to nonzero ones,

$$\int (a_{\mu}^{(n)}a_{\mu}^{(0)} + \phi_1^{(n)}\phi_1^{(0)} + \phi_2^{(n)}\phi_2^{(0)})d^2x = 0. \quad (3.13)$$

Using Expression (3.12) we have

$$\int \left[\frac{1}{e}(\partial_{\mu}\alpha)a_{\mu}^{(n)} + \alpha(\phi_1^{(n)}\varphi_2^{c1} - \phi_2^{(n)}\varphi_1^{c1}) \right] d^2x = 0. \quad (3.14)$$

Integrating by parts the first term in (3.14) and taking account of the arbitrariness of α , condition (3.11) follows. In order to treat the zero-mode problem, we shall follow the procedure described by Polyakov.⁷ We will adopt a special gauge condition (the "natural gauge") given by

$$\frac{1}{e}\partial_{\mu}A^{\mu} + \varphi_1^{c1}\varphi_2 - \varphi_2^{c1}\varphi_1 = 0 \quad (3.15)$$

and proceed then to eliminate the gauge freedom. We begin by calculating the Popov-Faddeev¹⁰ determinant in the natural gauge. We consider a gauge transformation

$$\delta A_{\mu} = \frac{1}{e}\partial_{\mu}\alpha,$$

$$\delta\varphi_1 = \alpha\varphi_2, \quad (3.16)$$

$$\delta\varphi_2 = -\alpha\varphi_1.$$

Then,

$$\delta \left(\frac{1}{e}\partial_{\mu}A^{\mu} + \varphi_1^{c1}\varphi_2 - \varphi_2^{c1}\varphi_1 \right) = \left(\frac{1}{e^2}\partial_{\mu}\partial^{\mu} - \varphi_1^{c1}\varphi_1 + \varphi_2^{c1}\varphi_2 \right) \alpha, \quad (3.17)$$

and the determinant is

$$\Delta = \det \left(\frac{1}{e^2}\nabla^2 + \varphi_1^{c1}\varphi_1 + \varphi_2^{c1}\varphi_2 \right). \quad (3.18)$$

Hence, the integration measure is given by

$$\text{measure} = \int DA_{\mu} D\varphi_1 D\varphi_2 \delta \left(\frac{1}{e}\partial_{\mu}A^{\mu} + \varphi_1^{c1}\varphi_2 - \varphi_2^{c1}\varphi_1 \right) \times \det \left(\frac{1}{e^2}\nabla^2 + \varphi_1^{c1}\varphi_1 + \varphi_2^{c1}\varphi_2 \right). \quad (3.19)$$

Let us now expand A_{μ} and φ using the complete set of solutions of Eqs. (3.9):

$$A_{\mu} = A_{\mu}^{c1} + \frac{1}{e}\partial_{\mu}\alpha + \sum_n \xi_n a_{\mu}^{(n)}, \quad (3.20)$$

$$\varphi = \varphi_{c1} + i\alpha\varphi_{c1} + \sum_n \xi_n \phi^{(n)}.$$

In order to express the integration measure in terms of $D\alpha$ and $d\xi_n$ we calculate the metric in the functional space of A_{μ} and φ , using the fact that the volume element corresponds to the square root of the determinant of the metric:

$$\begin{aligned} (\delta S)^2 &= \int d^2x [\delta A_{\mu}\delta A^{\mu} + (\delta\varphi_1)^2 + (\delta\varphi_2)^2] \\ &= \sum_n (d\xi_n)^2 + \int d\alpha \left(\frac{1}{e^2}\nabla^2 + |\varphi_{c1}|^2 \right) \delta\alpha d^2x. \end{aligned} \quad (3.21)$$

Then

$$DA_{\mu} D\varphi_1 D\varphi_2 = \prod_n d\xi_n D\alpha \det^{1/2} \left(\frac{1}{e^2}\nabla^2 + |\varphi_{c1}|^2 \right). \quad (3.22)$$

Substituting expansion (3.20) into expression (3.18), we can now integrate on α using the δ function. The answer is

$$\begin{aligned} \text{measure} = & \int \prod_n d\xi_n \det \left[\frac{1}{e^2} \nabla^2 + (\varphi_1^{c1} \varphi_1 + \varphi_2^{c1} \varphi_2) \right] \\ & \times \det^{-1/2} \left(\frac{1}{e^2} \nabla^2 + |\varphi_{c1}|^2 \right). \end{aligned} \quad (3.23)$$

We see that zero modes coming from gauge degrees of freedom are completely eliminated. But we still have zero modes associated with translation invariance. The corresponding eigenfunctions are

$$\begin{aligned} \tilde{a}_\mu^{(\lambda, 0)} &= \tilde{N}^{-1/2} \partial_\lambda A_\mu^{c1}, \\ \phi^{(\lambda, 0)} &= \tilde{N}^{-1/2} \partial_\lambda \varphi_{c1}, \end{aligned} \quad (3.24)$$

where $\tilde{N}$ is the normalization constant. Solutions (3.24) do not satisfy the natural gauge condition (3.15). But we can do an infinitesimal gauge transformation with a gauge function chosen in such a way that the new functions satisfy (3.15). The appropriate gauge transformation is

$$\alpha^{(\lambda)}(x) = -e A_\lambda^{c1}. \quad (3.25)$$

Now, we have

$$\begin{aligned} a_\mu^{(\lambda, 0)} &= N^{-1/2} (\partial_\lambda A_\mu^{c1} - \partial_\mu A_\lambda^{c1}) = F_{\lambda\mu}^{c1} N^{-1/2}, \\ \phi^{(\lambda, 0)} &= N^{-1/2} \nabla_\lambda^{c1} \varphi_{c1}, \end{aligned} \quad (3.26)$$

where

$$N = \int d^2x (F_{\mu\nu}^{c1} F^{\mu\nu c1} + |\nabla_\mu \varphi_{c1}|^2). \quad (3.27)$$

Condition (3.15) is now satisfied automatically since it reduces to the equation of motion for $F_{\mu\nu}^{c1}$.

If we call R_λ the center-of-mass coordinate of the vortex, it is easy to obtain its relation with the coordinate $\xi_0^{(\lambda)}$. Expanding $A_\mu^{c1}(x+R)$ we obtain

$$A_\mu^{c1}(x+R) = A_\mu^{c1}(x) + F_{\lambda\mu}^{c1} R^\lambda + \partial_\mu (A_\lambda R^\lambda). \quad (3.28)$$

Then, we can do the identification [using Eq. (3.16)]

$$\xi_0^{(\lambda)} = N^{1/2} R_\lambda \quad (3.29)$$

and the measure is

$$\text{measure} = N d\tilde{\mathbf{R}}. \quad (3.30)$$

Because in the one-loop approximation the difference between the two determinants in (3.23) can be neglected, we finally have

$$\text{measure} = N d\tilde{\mathbf{R}} \prod_{n \neq 0} e \xi_n \det^{1/2} \left(\frac{1}{e^2} \nabla^2 + \varphi_1^{c12} + \varphi_2^{c12} \right), \quad (3.31)$$

where n runs over nonzero modes only.

IV. VORTEX CONTRIBUTION TO FUNCTIONAL INTEGRALS

We are now in a position to evaluate the vortex contribution to the functional integral. Let us begin with the one-vortex contribution in the one-loop approximation. This is given from Eqs. (3.5), (3.6), (3.20), and (3.31):

$$Z_1 = \int d\tilde{\mathbf{R}} e^{-\epsilon^{(1)}} m_v^2 N \det^{1/2} \left(\frac{1}{e^2} \nabla^2 + |\varphi_{c1}|^2 \right) \exp \left(- \sum_{n \neq 0} \ln \frac{\Omega_n}{\Omega_0} \right), \quad (4.1)$$

$$Z_1 = \int d\tilde{\mathbf{R}} e^{-\epsilon^{(1)}} m_v^2 \beta(c_2, c_4; e^2), \quad (4.2)$$

where β is some function which can be in principle evaluated.

Hence, for widely separated vortices, from (4.2) and (2.30) we can express the contribution of the superposed vortices:

$$Z = \sum_{N, \{n_a\}} \frac{\mathcal{G}^N}{N!} \int \prod_{j=1}^N dR_j \exp \left[- \frac{\pi m_v^2}{e^2} \sum_{a \neq b} n_a n_b K_0(m_v |\tilde{\mathbf{R}}_a - \tilde{\mathbf{R}}_b|) \right]. \quad (4.3)$$

Here $\sum_{N, \{n_a\}}$ means a sum over all configurations $\{n_a\}$ with $n_a = \pm 1$, the $1/N!$ being included to eliminate double counting and

$$\mathcal{G} = m_v^2 \beta \exp(-\epsilon^{(1)}). \quad (4.4)$$

It will be useful for subsequent computations to have partition function (4.3) in a functional-integral representation.

In order to obtain such a representation we exploit the fact that

$$(\nabla^2 - m_v^2) K_0(m_v r) = -2\pi \delta^{(2)}(\tilde{\mathbf{r}}). \quad (4.5)$$

So, if we call $\Delta = K_0(m_v r)/2\pi$ we can rewrite the exponential in (4.3) as

$$\exp\left[-\frac{\pi m_v^2}{e^2} \sum_{a \neq b} n_a n_b K_0(m_v |\vec{R}_a - \vec{R}_b|)\right] = \exp\left[\frac{1}{2} \int \rho(\vec{x}) \Delta(\vec{x} - \vec{y}) \rho(\vec{y}) d^2x d^2y\right] \quad (4.6)$$

with

$$\rho(\vec{x}) = ib \sum_a n_a \delta^{(2)}(\vec{x} - \vec{R}_a), \quad (4.7)$$

$$b = 2\pi m_v / e. \quad (4.8)$$

But (4.8) can be easily written as a functional integral over a certain field ψ (Ref. 11):

$$\exp\left[\frac{1}{2} \int \rho(\vec{x}) \Delta(\vec{x} - \vec{y}) \rho(\vec{y}) d^2x d^2y\right] = \tilde{N} \int D\psi \exp\left\{-\frac{1}{2} \int [(\nabla\psi)^2 + m_v^2 \psi^2] d^2x\right\} \exp\left[\int \rho(\vec{x}) \psi(\vec{x}) d^2x\right]. \quad (4.9)$$

Using definition (4.8) we can integrate the argument in the second exponential of (4.9). Calling $b\psi = \chi$ we finally have

$$\exp\left[-\frac{\pi m_v^2}{e^2} \sum_{a \neq b} n_a n_b K_0(m_v |\vec{R}_a - \vec{R}_b|)\right] = N \int D\chi \exp\left\{-\frac{e^2}{8\pi^2 m_v^2} \int [(\nabla\chi)^2 + m_v^2 \chi^2] d^2x\right\} \exp\left[i \sum_a n_a \chi(\vec{R}_a)\right]. \quad (4.10)$$

Then, we have for Z

$$Z = N \int D\chi \exp\left\{-\frac{A}{2} \int [(\nabla\chi)^2 + m_v^2 \chi^2] d^2x\right\} \sum_{N, \{n_a\}} \int \prod_j d\vec{R}_j \frac{\xi^N}{N!} \exp\left[i \sum_a n_a \chi(\vec{R}_a)\right], \quad (4.11)$$

where $A = e^2 / 4\pi^2 m_v^2$.

We can now do the sum over configurations and rewrite the integrals on $d\vec{R}_j$ in the following way:

$$\begin{aligned} \sum_{N, \{n_a\}} \int \prod_j d\vec{R}_j \frac{\xi^N}{N!} \exp\left[i \sum_a n_a \chi(\vec{R}_a)\right] &= \sum_N \frac{\xi^N}{N!} \left[\int d\vec{R} (e^{i\chi(\vec{R})} + e^{-i\chi(\vec{R})}) \right]^N \\ &= \exp\left\{2\xi \int \cos[\chi(\vec{R})] d\vec{R}\right\}. \end{aligned} \quad (4.12)$$

Hence, we can write for Z

$$Z = N \int D\chi \exp\left\{-\frac{A}{2} \int [(\nabla\chi)^2 + m_v^2 \chi^2 - 2M^2 \cos(\chi)] d^2x\right\}, \quad (4.13)$$

where

$$M^2 = \frac{8\pi^2 m_v^2}{e^2} \mathcal{G} = \left(\frac{8\pi^2 m_v^2}{e^2} \beta\right) m_v^2 \exp(-\epsilon^{(1)}). \quad (4.14)$$

The effective coupling constant g of the χ field theory described by functional (4.13) can be estimated from the coefficient of χ^4 in the expansion of $\cos\chi$:

$$g \sim \frac{M^2}{Ae^2} \sim \exp(-\epsilon^{(1)}) = \exp\left(-\frac{m_v^2}{e^2} Q\right). \quad (4.15)$$

That is, the effective coupling g is exponentially small.

We can easily evaluate correlation functions if we introduce the generating functional

$$\left\langle \exp\left[i \int \rho(x) \eta(x) d^2x\right] \right\rangle = \frac{Z[\eta]}{Z[0]}, \quad (4.16)$$

where

$$\rho(\vec{x}) = \sum_a n_a \delta(\vec{x} - \vec{x}_a) \quad (4.17)$$

and

$$Z[\eta] = N \int D\chi \exp\left(-\frac{A}{2} \int \{[\nabla(\chi - \eta)]^2 + m_v^2 (\chi - \eta)^2 - 2M^2 \cos\chi\} d^2x\right) \quad (4.18)$$

is obtained following the same procedure explained for Z [Eq. (4.13)]. Then, we have

$$\langle \rho(x) \rho(y) \rangle = -\frac{1}{Z[0]} \frac{\delta^2 Z[\eta]}{\delta \eta(x) \delta \eta(y)} \quad (4.19)$$

and, defining as usual Fourier transforms

$$\langle \rho(x) \rho(0) \rangle = \int \frac{d^2k}{(2\pi)^2} e^{-i\vec{k} \cdot \vec{x}} \langle \rho(k) \rho(-k) \rangle, \quad (4.20)$$

we obtain

$$\begin{aligned} \langle \rho(k) \rho(-k) \rangle &= A[(k^2 + m_v^2) \\ &\quad - A(k^2 + m_v^2) \langle \chi(k) \chi(-k) \rangle]. \end{aligned} \quad (4.21)$$

We will use this result to obtain correlation functions for the "magnetic" field, which was given, classically by expression (2.26):

$$H = \frac{m_v^2}{e} \sum_a n_a K_0(m_v |\vec{r} - \vec{r}_a|).$$

The Fourier transform of the magnetic field is related to $\rho(k)$ in the following way:

$$H(k) = -\frac{2\pi m_v^2}{e} \frac{1}{k^2 + m_v^2} \rho(k). \quad (4.22)$$

Then, we have

$$\begin{aligned} \langle H(k)H(-k) \rangle &= m_v^2 \left(\frac{1}{k^2 + m_v^2} - A \langle \chi(k)\chi(-k) \rangle \right) \\ &+ \langle H(k)H(-k) \rangle^{(0)}, \end{aligned} \quad (4.23)$$

where $\langle H(k)H(-k) \rangle^{(0)}$ represents the no-vortex contribution to the correlation function. Because of our previous comment about the exponentially small coupling of the χ field, we can write

$$A \langle \chi(k)\chi(-k) \rangle = \frac{1}{k^2 + m_v^2 + M^2}, \quad (4.24)$$

and we finally obtain

$$H(k)H(-k) = 1 - \frac{m_v^2}{k^2 + m_v^2 + M^2}. \quad (4.25)$$

We see that the singularity at $-k^2 = m_v^2 + M^2$ depends on the pseudoparticle contribution through the term $M^2 \propto m_v^2 \exp[-(m_v^2/e^2)Q]$.

V. CONFINEMENT

We will now discuss the theory developed in previous sections when heavy charged particles are introduced. To study the possibility of confinement of these particles due to pseudoparticle contributions, we will use the correlation function introduced by Wilson¹² to calculate the transition amplitude of a process in which two heavy particles of charge $\pm q$ are created, then separated by a distance R for a long time T , and finally annihilate. The Wilson correlation function reads

$$F[C] = \left\langle \exp \left(i q \oint_C A_\mu dx^\mu \right) \right\rangle, \quad (5.1)$$

$$F[C] = \frac{\int D\chi \exp \left(-\frac{1}{2} A \int \{ [\nabla(\chi - \eta)]^2 + m_v^2 (\chi - \eta)^2 - 2M^2 \cos \chi \} d^2x \right)}{\int D\tilde{\chi} \exp \left(-\frac{1}{2} A \int \{ (\nabla\tilde{\chi})^2 + m_v^2 \tilde{\chi}^2 - 2M^2 \cos \tilde{\chi} \} d^2x \right)}. \quad (5.7)$$

The most important contribution to (5.7) comes from those values of $\chi, \tilde{\chi}$ satisfying the classical equations

$$\nabla^2(\chi_{cl} - \eta) = M^2 \sin \chi_{cl} + m_v^2(\chi_{cl} - \eta), \quad (5.8)$$

$$\nabla^2 \tilde{\chi}_{cl} = M^2 \sin \tilde{\chi}_{cl} + m_v^2 \tilde{\chi}_{cl}, \quad (5.9)$$

because from the previous discussion [see Eq. (4.15)] the effective coupling of the theory is exponentially

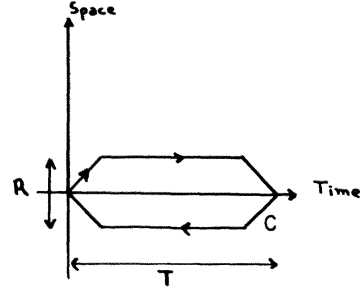


FIG. 1. The loop diagrams represents two heavy particles of opposite charges which are created, then separated by a distance R , remain there for a long time T and finally annihilate.

where C is the contour of the loop represented in Fig. 1.

Because $F[C] = \exp[E(R)T]$, the calculation of the right-hand side in expression (5.1) permits us to evaluate the interaction energy between the two particles. In order to do this, let us use the fact that classically

$$q \oint A_\mu dx^\mu = q \int_{S_C} H(\vec{r}) d\vec{r}, \quad (5.2)$$

where S_C is the area enclosed by the loop.

But from Eq. (4.22) we have that classically

$$H(\vec{r}_1) = \frac{m_v^2}{e} \int \rho(\vec{r}_2) K_0(m_v |\vec{r}_1 - \vec{r}_2|) d\vec{r}_2. \quad (5.3)$$

Then

$$q \oint A_\mu dx^\mu = \int d\vec{r}_2 \rho(\vec{r}_2) \eta(\vec{r}_2), \quad (5.4)$$

where

$$\eta(\vec{r}_2) = \frac{q m_v^2}{e} \int_{S_C} K_0(m_v |\vec{r}_1 - \vec{r}_2|) d\vec{r}_1. \quad (5.5)$$

Then, we have for $F[C]$

$$F[C] = \left\langle \exp \left(i \int \rho(x) \eta(x) d^2x \right) \right\rangle, \quad (5.6)$$

and we can use expression (4.16) to evaluate it:

small and fluctuation corrections to the field χ_{cl} are negligible.

From the definition of η we have

$$\nabla^2 \eta - m_v^2 \eta = \frac{2\pi q}{e} \Theta(S_C), \quad (5.10)$$

where

$$\Theta(S_C) = \begin{cases} 1 & \text{if } \vec{x} \in S_C \\ 0 & \text{if } \vec{x} \notin S_C. \end{cases} \quad (5.11)$$

We can then rewrite Eqs. (5.8), (5.9)

$$\nabla^2 \chi_{cl} = M^2 \sin \chi_{cl} + m_v^2 \chi_{cl} + \frac{2\pi q}{e} m_v^2 \Theta(S_C), \quad (5.12)$$

$$\nabla^2 \tilde{\chi}_{cl} = M^2 \sin \tilde{\chi}_{cl} + m_v^2 \tilde{\chi}_{cl} \quad (5.13)$$

and, for $F[C]$,

$$F[C] = \frac{\exp\{-\frac{1}{2}A \int [-(\chi_{cl} - \eta) \nabla^2 (\chi_{cl} - \eta) + m_v^2 (\chi_{cl} - \eta)^2 - 2M^2 \cos \chi_{cl}] d^2x\}}{\exp\{-\frac{1}{2}A \int (\tilde{\chi}_{cl} \nabla^2 \tilde{\chi}_{cl} + m_v^2 \tilde{\chi}_{cl}^2 - 2M^2 \cos \tilde{\chi}_{cl}) d^2x\}}, \quad (5.14)$$

$$F[C] = \frac{\exp(-\frac{1}{2}AI_N)}{\exp(-\frac{1}{2}AI_D)}. \quad (5.15)$$

We can rearrange the integral in the numerator in the following form:

$$I_N = I_N^1 + I_N^2 \quad (5.16)$$

with

$$I_N^1 = \int d^2x (-\chi_{cl} \nabla^2 \chi_{cl} + m_v^2 \chi_{cl}^2 - 2M^2 \cos \chi_{cl}), \quad (5.17)$$

$$I_N^2 = \int d^2x \left\{ \eta \left[\nabla^2 \chi_{cl} - m_v^2 \chi_{cl} - \frac{2\pi q}{e} m_v^2 \Theta(S_C) \right] + \frac{2\pi q}{e} m_v^2 \Theta(S_C) \right\}. \quad (5.18)$$

We see that I_N^1 has formally the same form as I_D . Since outside the loop χ_{cl} and $\tilde{\chi}_{cl}$ obey the same equation, their contribution outside the loop cancels. Hence,

$$F[C] = \frac{\exp[-\frac{1}{2}A \int_{S_C} (-\chi_{cl} \nabla^2 \chi_{cl} + m_v^2 \chi_{cl}^2 - 2M^2 \cos \chi_{cl}) d^2x] \exp(-\frac{1}{2}AI_N^2)}{\exp[-\frac{1}{2}A \int_{S_C} (-\tilde{\chi}_{cl} \nabla^2 \tilde{\chi}_{cl} + m_v^2 \tilde{\chi}_{cl}^2 - 2M^2 \cos \tilde{\chi}_{cl}) d^2x]}. \quad (5.19)$$

The most important contribution to the denominator comes from constant solutions of (5.13), that is $\tilde{\chi}_{cl} = 0$. In this case,

$$\exp\left[-\frac{A}{2} \int_{S_C} (-\tilde{\chi}_{cl} \nabla^2 \tilde{\chi}_{cl} + m_v^2 \tilde{\chi}_{cl}^2 - 2M^2 \cos \tilde{\chi}_{cl}) d^2x\right] = \exp(AS_C M^2), \quad (5.20)$$

where S_C is the area enclosed by the loop.

Outside the loop $\chi_{cl} = 0$ is also a solution giving a vanishing contribution to the integral I_N^2 . We thus have

$$F[C] = \exp(-AS_C M^2) \exp\left[-\frac{A}{2} \int_{S_C} (-\chi_{cl} \nabla^2 \chi_{cl} + m_v^2 \chi_{cl}^2 - 2M^2 \cos \chi_{cl}) d^2x\right] \\ \times \exp\left\{-\frac{A}{2} \int_{S_C} \left[\eta \left(\nabla^2 \chi_{cl} - m_v^2 \chi_{cl} - \frac{2\pi q m_v^2}{e}\right) + \frac{2\pi q m_v^2 \chi}{e}\right] d^2x\right\} \quad (5.21)$$

with

$$\nabla^2 \chi_{cl} = M^2 \sin \chi_{cl} + m_v^2 \left(\chi_{cl} + \frac{2\pi q}{e}\right). \quad (5.22)$$

We see that if $q/e = \text{integer}$, there is only one constant solution $\chi = -2\pi q/e = -2\pi k$. This solution

gives the most important contribution to the integral. In that case

$$F[C]_{q/e=k} = \exp\{-AM^2 S_C - \frac{1}{2}AS_C[m_v^2(2k\pi)^2 - 2M^2] \\ + \frac{1}{2}AS_C m_v^2(2k\pi)^2\}, \quad (5.23)$$

$$F[C]_{q/e=k} = 1.$$

Then, there is no confinement in the sense of Wilson.

But, if we take $q/e \neq \text{integer}$, the result is quite different. For example, for $q/e = \frac{1}{2}$ we obtain

$$F[C] = \exp(-2AM^2RT), \quad (5.24)$$

where we have put $S_C = RT$. Hence

$$\epsilon(R) = 2AM^2R = 4m_V^2\beta \exp(-\epsilon^{(1)})R, \quad (5.25)$$

that is,

$$\epsilon(R) \propto m_V^2 \exp\left(-\frac{m_V^2}{e^2}Q\right)R. \quad (5.26)$$

Thus, fractionally charged quarks are confined, a result advanced in this model by the Callan-Dashen-Gross calculation in the dilute-gas approximation. It is easy to see that for $q/e \neq \text{integer}$ (not necessarily semi-integer) $\ln F[C]$ is proportional to S_C , thus giving confinement.

This resembles the results found by Coleman, Jackiw, and Susskind¹³ in the massive Schwinger model: When an external current (corresponding to a charge distribution consisting of two opposite charges of magnitude q) is coupled to the model, there are no long-range forces whenever q is an integral multiple of e . On the contrary, for arbitrary q the long-range force is present. From this Coleman *et al.* infer that the Higgs phenomenon does not occur. In the case of the Nielsen-Olesen model, one can think that pseudoparticle contributions restore the symmetry, the Higgs phenomenon does not occur, and, as in the massive Schwinger model, only fractional external charges (which cannot bind to charged scalars) will be confined.

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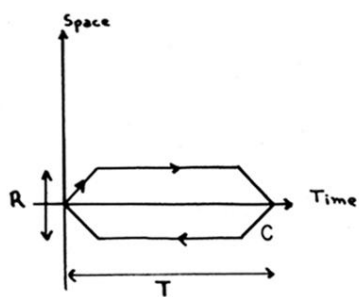


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